

Conceptual Design Report

4) Structures

4.1 Requirements and Constraints:

- a) To provide a structural framework that would withhold the stresses and forces for all the conditions during the mission.
- b) The wingspan has to be a minimum of 3 feet.
- c) The payload has to be placed inside the flying wing hence the structure should have enough room (thickness).
- d) The aircraft should be made as light as possible without neglecting the structural abilities and integrity.

4.2 Assumptions:

- a) The beams used in the structure are assumed to be prismatic, homogeneous and uniform.
- b) Other properties like thermal, impact etc. are ignored at this phase of design.
- c) The materials used are assumed to be Hookean in nature.
- d) Half the wing accounts for 50% of the total lift.
- e) The lift generated is uniform across the wingspan and the lift profile is rectangular.

4.3 Material Selection and Properties: A careful analysis of different materials was done for their physical and structural properties. Balsa and spruce were chosen because of their high Modulus of Elasticity and high strength to weight ratio. These two materials are easily available and cheap too.

4.4 Factors of Safety and Load Factors^M:

- a) **Limit Load:** The maximum load the aircraft is going to experience during the flight. The load is typically the highest amount of lift generated.
- b) **Proof Load:** The product of Limit load and proof factor gives us the Proof load which in reality is the amount of load the structure should hold without any damage. The proof factor lies somewhere between 1.0-1.5.
- c) **Ultimate Load:** The product of Limit Load and the ultimate factor gives us the ultimate load, which is the load beyond which the structure fails. The ultimate factor is usually 1.5..

4.5 Beam analysis^N:

Euler-Bernoulli's beam theory was used to analyze the different cross sections of beams. Two major spars will be utilized at approximately 25% and 70% chord. The center of pressure is more likely to be along the 25% chord line and hence the main spar (at 25%) would account for approximately 75% of the wing loading. The aft-spar at 70% chord would carry the remaining 25% load and also support control surfaces like ailerons. The lift distribution was assumed to be rectangular and uniform across the wingspan, which is actually the worst case scenario.

- a) **I-Beam:** I-beams for different materials were analyzed to carry the proof load without significant bending; I-beams weigh lesser and provide significant moment of Inertia. History tells us that I-beams are one of the best choices when it comes to spars for aircraft wings. The maximum deflection at the wing tip while experiencing the proof load was found to be about **1.3** inches.
- b) **Solid Rectangular Beam:** Solid rectangular beams for different materials were analyzed for the spars. These beams provide good structure and rigidity but weigh heavier than the I-Beams.
- c) **Circular beams for the booms:** Circular beams will be used to build the twin-booms. The booms will have to restrain bending as well as torsional loads.

4.6 Stresses on the Beams:

- a) **Shear Stress**^O: Shear stresses were calculated along the beam using the Mechanics of Material's approach. The maximum shear stress was observed at the midsection of the wing. Figure 4.2 shows the distribution of shear stress along the wing span. The maximum shear stress was calculated to be **95 Psi**. (Fig 4.3)
- b) **Internal Moments**: Internal moments were calculated along the beam using the Mechanics of Material's approach.
- c) **Normal Stress (Bending)**^P: Normal stress or bending stress was calculated along the wing-span, again the maximum normal stress was found to be at the center of the wing-body.

The maximum Bending stress calculated was about **3.6 Ksi** (Fig. 4.3)

4.7 Twin Booms: There are 2 booms that connect to the power plant and all the way across to the tail section. The vertical stabilizers are attached to the booms. The booms will provide structural stability and support for the airframe. They will also act as the moment carriers for stability and control. The twin booms will be made off thin walled tubular beams for structural rigidity and strength.

4.8 Landing Gears: For the purpose of ground control the best type of landing gear would be a tricycle for the flying wing. The gears will be picked off the shelf, readily available in the market and the landing gear assembly will be designed such that it would be capable of handling 5g loads (Proof Load).

4.9 Tails: Tails (Vertical and Horizontal) will use spars made up of spruce and have a I cross-section. Further analysis on the tails will be done at a later phase of design.

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4.10 Material Finalized:

- a) Spars – Spruce sticks
- b) Ribs – 1/16th and 1/8th Balsa
- c) Aluminum 6061T for the Tail-Booms.
- d) Bulkheads: Balsa and spruce sticks.

4.11 Spars: - Two spars (main and aft) will be used to make the wing structurally sound. Because of the uneven geometry, placing the main spar at 25% of the MAC would involve using joints, which would mean weaker points and would also complicate the structural analysis. The main spar would be at 25% at the wing tips and the center section. The center of the aircraft would have stronger ribs to transfer and account for the moment generated by the forces at the AC. The spars are designed to withhold bending stresses up to 3.8Ksi^{4.1} and shear stress of 164 Psi, 107 Psi at the main and aft spars respectively. The detailed dimensions and stresses on the spars can be found on figure 4.3 and 4.4 respectively^{4.3}.

4.12 Ribs: Ribs are responsible to give the preferred aerodynamic shape to the wing-body. After careful analysis and shear force calculations balsa wood sheets was found to be the best choice for the ribs. Two different thicknesses of ribs will be used along the span. The payload section (15'x24') will use 1/8th balsa whereas the rest of the wing will use 1/16th Balsa. Shear flows and shear stresses were calculated along the surface of the ribs to make sure that they can handle the different forces generated during the flight.

4.13 Twin Booms: Twin booms on the aircraft prove to be very vital as they connect most of the important parts of the airplane. The booms run from the very tip of the aircraft till the very end. The motors will be mounted on the booms and the booms would carry through the wings and into the tails as shown in the figure 5.1. Finding a good material for the boom was crucial as it is almost 4 feet long and we have 2 of them. After careful analysis thin-walled cylindrical aluminum tubes were found to be strong and light enough to hold all the bending and twisting. The booms selected were of 3/8th inch diameter with 0.035' thickness. The 2 booms combined weigh about 0.3 Lbs...

4.14 Payload Section: The tennis balls will be placed between the ribs somewhere along the center-line of the aircraft. Span-loading is incorporated to reduce weight and reduce structural complexity. The bombing mechanism is discussed at a later part of the report. Thin sheets of ply would be used as the bomb-bay door. They will be hinged to the ribs and spars. A small scaled prototype of the bomb bay and the doors was analyzed to predict the thickness of the ply and the amount of stresses it encounters.

4.15 Tails (Horizontal and vertical): The actual sizing and location of the tails can be found in table 6.1. Two verticals will have 1 spar located at the aft to support the rudder and provide a better structure. 1/16th Balsa will be used for ribs and ½ x ½ (inches) spruce sticks will be used for spars. Similarly the horizontal will have a spar to support the elevator as well as provide the strength to account for the lift and moments on the tails.

4.16 Landing Gear: The landing-gear configuration finalized for the aircraft is a tail-dragger. 2 main gears and two tail-draggers would be used for take-offs and landings. The two main gears would hold almost 90% of the aircraft's weight. The landing gear has been analyzed to withhold loads 4g loads. Wheels used for the landing gear will be off-the-shelf foam wheels. The tail-draggers will be connected to the rudder via thin aluminum/steel rods.

Detail Design:

4.17 Ribs & Spars: The ribs were made up of 1/8th and 1/16th balsa as analyzed during the preliminary design phase. The spars had to be modified slightly in order for them to be readily available in the market. Precautions were taken while choosing the sizes keeping in mind the load factors and stresses associated with them. The final sizings of the spars are provided on Figure 4.4. The overall weight of the spars increased approximately by 7% but that was counterbalanced by making lightening holes on the ribs and the landing gears.

4.18 Booms: The aluminum booms were sized exactly as analyzed during the preliminary design phase. No changes were made as they were readily available in the market. The booms were attached to the motors on one end and the tails at the other end. The bending on the boom was approximately 1.2E-3 inches^{4.2} at the load factor of 3.8.

4.19 Landing Gears: Landing gears were purchased off the shelf in accordance with the analysis done during the preliminary design. Aluminum plates and tubes were used to make the landing gear mounting plates which were then welded to the booms.

Ground testing and validation:

Experimental Results: Structural load test was performed by loading the wing along the spar with sand-bags. 19 sand bags each weighing 1.02 lbs. were loaded uniformly along the main-spar and 9 sand-bags were loaded uniformly along the aft spar each weighing 0.52 lbs. The total load on the wing was 24 lbs. The wing deflected by 1.13 inches, approximately 0.07 inches lesser than what was predicted.

Discussions: As far as the slight deviation on the spars' deflection is concerned, it's mainly due to not so accurate loading. Accurate loading was not possible because of ribs in between the spars. In addition to that the spars used on the aircraft were slightly bigger than the one initially analyzed, that was to avoid construction issues (Joints) as these were off the shelf readily available spar sticks. The spars are still within the estimated bending region (75% of fracture state). As a conclusion the structure will be able to hold the desired load factors and different flight conditions during the mission.

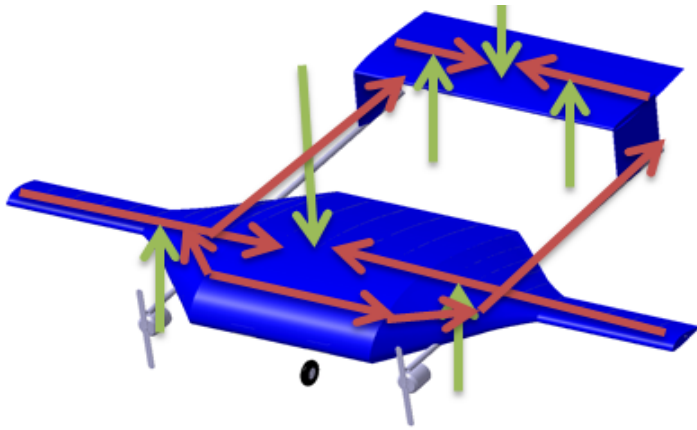


Fig:4.1 Major Load Paths on the aircraft.

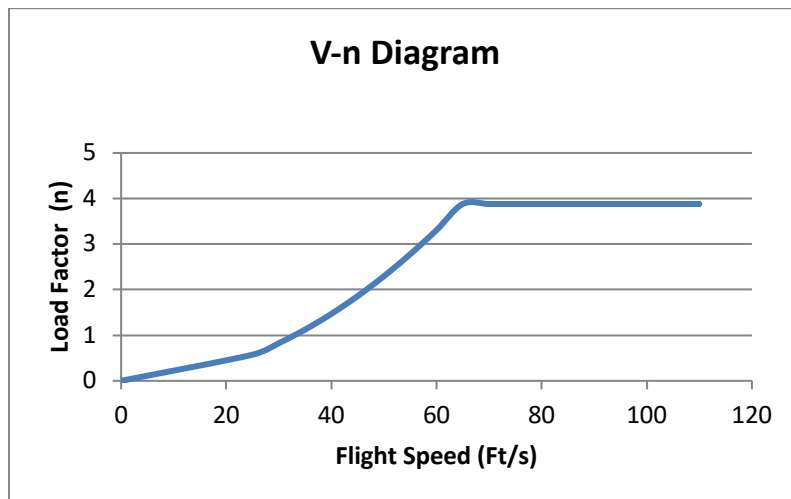


Fig: 4.2 V-n Diagram and load factor estimations.

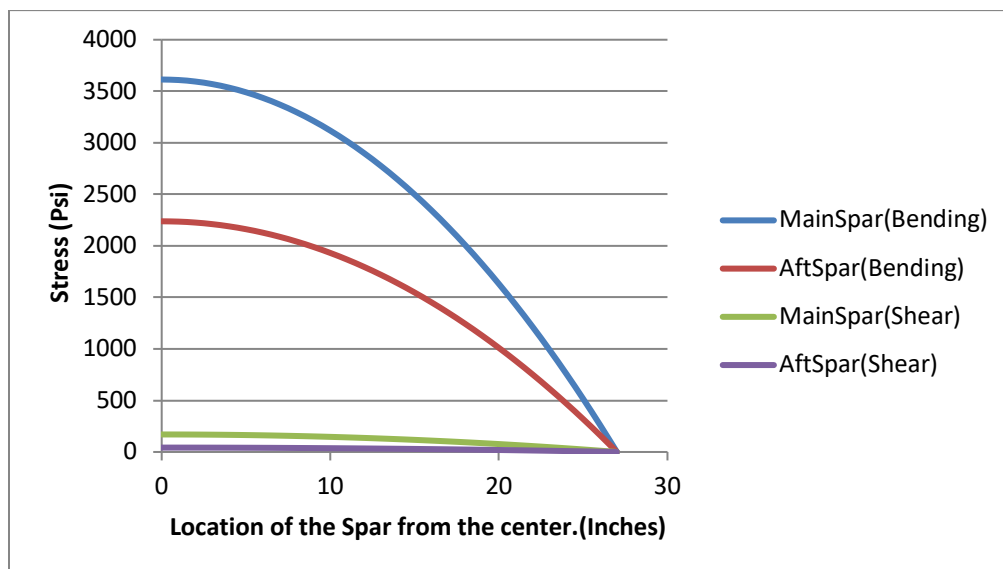


Fig: 4.3 Bending and shear stress distribution on the spars.

Bending along the span at the maximum load situation.						Main SPAR		
Section Properties								
Material		I-Beam	Area(sq in.)	Volume	Weight	MOI(x)	MOI(y)	Deflection(in)
Spruce	b	0.7500	0.3125	16.8750	0.2700	0.0212	0.0043	1.1259
	h	0.5000						
	H	0.7500						
	t	0.2500						
Bending along the span at the maximum load situation.						Aft Spar		
Section Properties								
Material		I-Beam	Area(sq in.)	Volume	Weight	MOI(x)	MOI(y)	Deflection(in)
Spruce	b	0.4375	0.1484375	8.015625	0.12825	0.005694		1.045958691
Balsa	h	0.3125						
	H	0.5625						
	t	0.125						

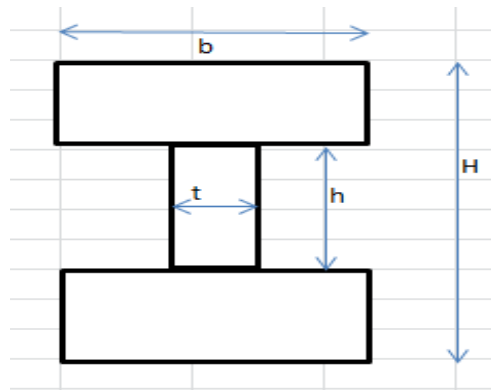


Fig 4.4 Section properties and deflections on the main and aft spars.

References.

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